

Introduction to Probabilistic Machine Learning

Gaussian Processes

Ralf Herbrich

Overview

1. Basic Concepts
2. Gaussian Processes for Regression
3. Weight-Space View of Gaussian Processes Regression
4. Gaussian Processes for Classification

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Unit 10 – Gaussian Processes

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Distribution over Functions

- **Status:** We assumed that we have a set of *parameterized* functions $f(x; \mathbf{w})$ and the distribution over functions f comes from a distribution over $\mathbf{w} \in \mathbb{R}$
 - Required *explicit* definition of M basis functions $\phi: \mathcal{X} \rightarrow \mathbb{R}$
 - The most expensive operation was $O(M^3)$ (matrix inversion)
- **Question:** How would a *distribution over functions* f be defined if we do not have explicit parameterized functions?
- **Observations:** A function is a mapping from an input space $\mathcal{X}$ to real values $\mathbb{R}$
- **Idea:** A distribution over functions can be specified by specifying the distribution of function values $f(x_1), \dots, f(x_n)$ for *any* $x_1, \dots, x_n \in \mathcal{X}$!
 - The higher the correlation of function values $f(x_i)$ and $f(x_j)$ of nearby inputs x_i and x_j , the smoother the function f

$$f(x; \mathbf{w}) = \mathbf{w}^T \boldsymbol{\phi}(x)$$

weight vector feature vector

- **Gaussian Process.** A Gaussian process is a probability distribution over functions $f: \mathcal{X} \rightarrow \mathbb{R}$ that has the property that for any $x_1, \dots, x_n \in \mathcal{X}$

$$p\left(\begin{bmatrix} f(x_1) \\ \vdots \\ f(x_n) \end{bmatrix}\right) = \mathcal{N}\left(\begin{bmatrix} f(x_1) \\ \vdots \\ f(x_n) \end{bmatrix}; \begin{bmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{bmatrix}, \begin{bmatrix} C(x_1, x_1) & \dots & C(x_1, x_n) \\ \vdots & \ddots & \vdots \\ C(x_n, x_1) & \dots & C(x_n, x_n) \end{bmatrix}\right)$$

- Fully parameterized by a mean *function* $m: \mathcal{X} \rightarrow \mathbb{R}$ and covariance *function* $C: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$
- Mean function is arbitrary, but the covariance function must be positive definite!

- **Gaussian processes** have a long history in engineering ("Kriging")

- In 1950, Danie G. Krige was studying gold locations predicted from a few boreholes
- In 1960, Georges Matheron rediscovered Krige's work and formalized it for statistics
- In 1999, Prof. David MacKay rediscovered Gaussian Processes for machine learning



Danie G. Krige
(1919 – 2013)



Georges Matheron
(1930 – 2000)



Sir David JC MacKay
(1967 – 2016)

Sampling Multivariate Gaussians

■ **Question:** How do we sample $\mathbf{F} \sim \mathcal{N}(\cdot; \boldsymbol{\mu}, \boldsymbol{\Sigma})$ for arbitrary $\boldsymbol{\mu} \in \mathbb{R}^n$, $\boldsymbol{\Sigma} \in \mathbb{R}^{n \times n}$?

1. Compute the Cholesky decomposition $\mathbf{L}$ of the matrix $\boldsymbol{\Sigma}$ such that $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^T$
2. Sample n uniform random numbers $u_1, \dots, u_n$ between 0 and 1 and use Box-Mueller to transform into n standard normal variables $r_1, \dots, r_n \sim \mathcal{N}(\cdot; \mathbf{0}, \mathbf{I})$ via

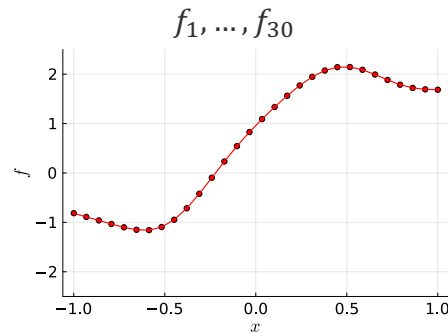
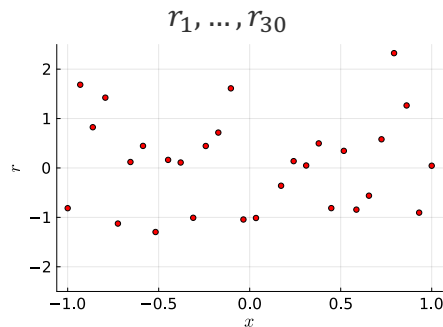
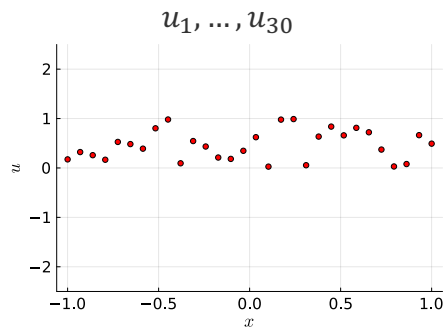
$$\begin{bmatrix} r_{2i-1} \\ r_{2i} \end{bmatrix} = \begin{bmatrix} \sqrt{-2 \ln(u_{2i-1})} \cdot \cos(2\pi \cdot u_{2i}) \\ \sqrt{-2 \ln(u_{2i-1})} \cdot \sin(2\pi \cdot u_{2i}) \end{bmatrix}$$

3. Compute a complete sample of the function values $\mathbf{f}$ using

$$\mathbf{f} = \mathbf{L}\mathbf{r} + \boldsymbol{\mu}$$

$$\boldsymbol{\mu} = \mathbf{0}$$

$$\Sigma_{i,j} = \exp(-4(x_i - x_j)^2)$$



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Gaussian Process: Example

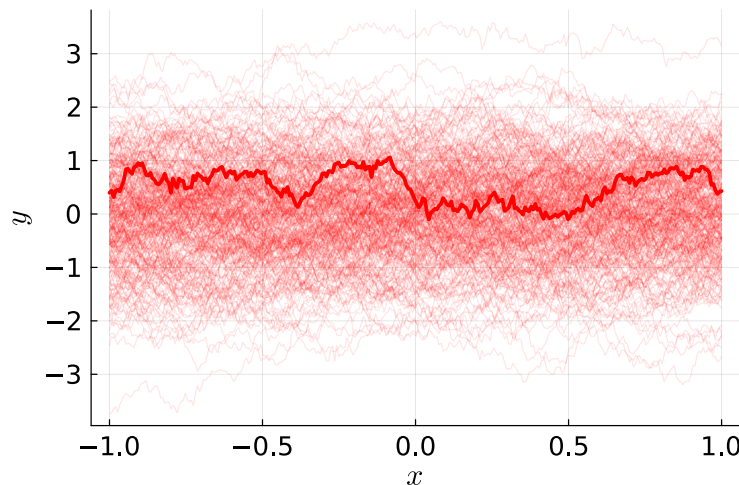
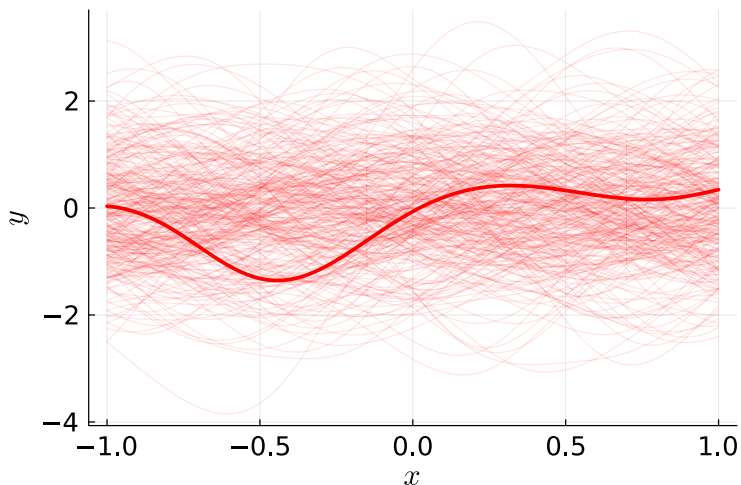
$$m(x) = 0$$

RBF Covariance Function

$$C(x, x') = \exp\left(-\frac{(x - x')^2}{\lambda^2}\right)$$

Ornstein-Uhlenbeck Covariance Function

$$C(x, x') = \exp\left(-\frac{|x - x'|}{\lambda}\right)$$



Overview

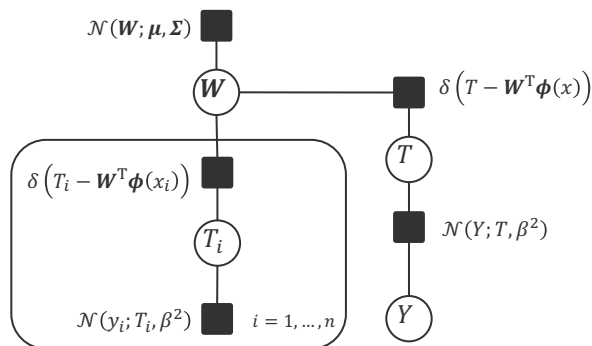
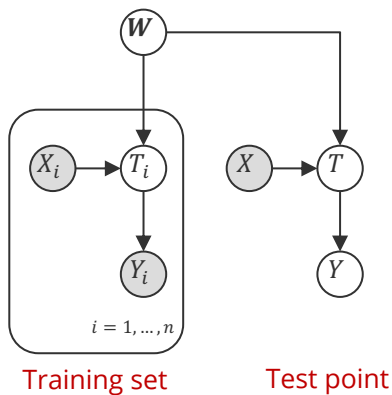
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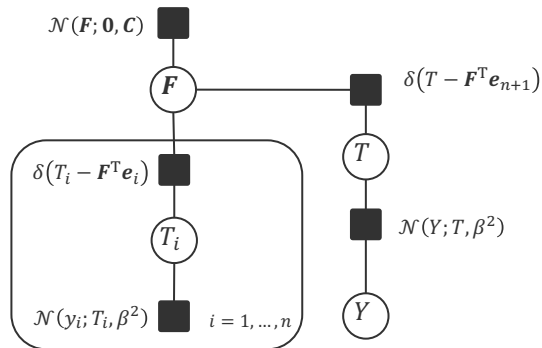
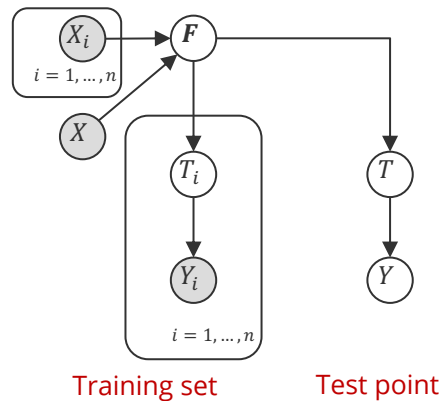
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Gaussian Process Regression as Factor Graph

Bayesian Linear Regression

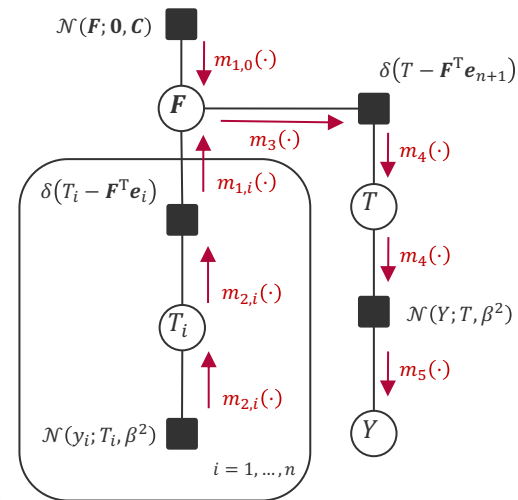


Gaussian Process Regression



Gaussian Process Regression via Message Passing

- For each training example we assume Gaussian noise:
 - $m_{2,i}(t_i) = \mathcal{N}(t_i; y_i, \beta^2)$
- The distribution over the $n + 1$ function values $\mathbf{f}$ is the product of
 - GP prior: $m_{1,0}(\mathbf{f}) = \mathcal{N}(\mathbf{f}; \mathbf{0}, \mathbf{C})$
 - Training data messages: $m_{1,i}(\mathbf{f}) \propto \mathcal{G}(\mathbf{f}; \beta^{-2} y_i \cdot \mathbf{e}_i, \beta^{-2} \cdot \mathbf{e}_i \mathbf{e}_i^T)$
- Hence, the message from the $n + 1$ function values $\mathbf{F}$ is Gaussian!
 - GP posterior: $m_3(\mathbf{f}) = \mathcal{N}(\mathbf{f}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \mathcal{N}\left(\begin{bmatrix} \mathbf{f}_n \\ \mathbf{f}_{n+1} \end{bmatrix}; \begin{bmatrix} \boldsymbol{\mu}_n \\ \boldsymbol{\mu}_{n+1} \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_n & \boldsymbol{\sigma}_n \\ \boldsymbol{\sigma}_n^T & \boldsymbol{\Sigma}_{n+1,n+1} \end{bmatrix}\right)$
- Using the Gaussian projection factor, the message to the latent function value T and prediction Y at the test point x is also Gaussian!
 - Test point function value: $m_4(t) = \mathcal{N}(t; \boldsymbol{\mu}^T \mathbf{e}_{n+1}, \mathbf{e}_{n+1}^T \boldsymbol{\Sigma} \mathbf{e}_{n+1}) = \mathcal{N}(t; \boldsymbol{\mu}_{n+1}, \boldsymbol{\Sigma}_{n+1,n+1})$
 - Prediction function value: $m_5(y) = \mathcal{N}(y; \boldsymbol{\mu}_{n+1}, \boldsymbol{\Sigma}_{n+1,n+1} + \beta^2)$
- We need to compute final row of the partitioned mean and covariance!



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Mixed Multivariate Gaussian Multiplication

GP Posterior

$$m_3(\mathbf{f}) = \mathcal{N}(\mathbf{f}; \mathbf{0}, \mathbf{C}) \cdot \mathcal{G}\left(\mathbf{f}; \beta^{-2} \sum_{i=1}^n y_i \cdot \mathbf{e}_i, \beta^{-2} \sum_{i=1}^n \mathbf{e}_i \mathbf{e}_i^T\right)$$

GP prior Data likelihood

- Theorem (Mixed Multiplication).** Given two multi-dimensional Gaussian distributions $\mathcal{N}(\mathbf{x}; \mathbf{m}, \mathbf{S})$ and $\mathcal{G}(\mathbf{x}; \boldsymbol{\tau}, \mathbf{P})$ we have

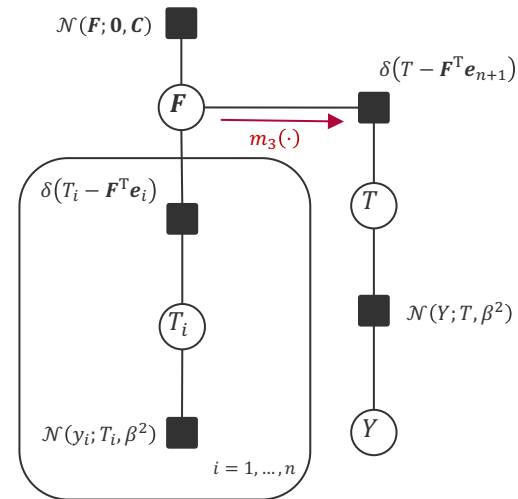
$$\mathcal{N}(\mathbf{x}; \mathbf{m}, \mathbf{S}) \cdot \mathcal{G}(\mathbf{x}; \boldsymbol{\tau}, \mathbf{P}) \propto \mathcal{N}(\mathbf{x}; (\mathbf{I} + \mathbf{S}\mathbf{P})^{-1}(\mathbf{m} + \mathbf{S}\boldsymbol{\tau}), (\mathbf{I} + \mathbf{S}\mathbf{P})^{-1}\mathbf{S})$$

Partitioned GP Posterior

$$m_3(\mathbf{f}) = \mathcal{N}\left(\begin{bmatrix} \mathbf{f}_n \\ \mathbf{f}_{n+1} \end{bmatrix}; \begin{bmatrix} \boldsymbol{\mu}_n \\ \boldsymbol{\mu}_{n+1} \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_n & \boldsymbol{\sigma}_n \\ \boldsymbol{\sigma}_n^T & \boldsymbol{\Sigma}_{n+1,n+1} \end{bmatrix}\right)$$

$$\left(\begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix} + \begin{bmatrix} \mathbf{K} & \mathbf{k} \\ \mathbf{k}^T & k \end{bmatrix} \begin{bmatrix} \beta^{-2} \mathbf{I} & \mathbf{0} \\ \mathbf{0}^T & 0 \end{bmatrix} \right)^{-1} \begin{bmatrix} \mathbf{K} & \mathbf{k} \\ \mathbf{k}^T & k \end{bmatrix} \begin{bmatrix} \beta^{-2} \mathbf{y} \\ 0 \end{bmatrix}$$

$$\left(\begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix} + \begin{bmatrix} \mathbf{K} & \mathbf{k} \\ \mathbf{k}^T & k \end{bmatrix} \begin{bmatrix} \beta^{-2} \mathbf{I} & \mathbf{0} \\ \mathbf{0}^T & 0 \end{bmatrix} \right)^{-1} \begin{bmatrix} \mathbf{K} & \mathbf{k} \\ \mathbf{k}^T & k \end{bmatrix}$$



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Inverse of a Partitioned Matrix

■ Key Inverse to Compute

$$\left(\begin{bmatrix} I & 0 \\ 0^T & 1 \end{bmatrix} + \begin{bmatrix} K & k \\ k^T & \beta^2 \end{bmatrix} \begin{bmatrix} \beta^{-2} I & 0 \\ 0^T & 0 \end{bmatrix} \right)^{-1} = \begin{bmatrix} I + \beta^{-2} K & 0 \\ \beta^{-2} k^T & 1 \end{bmatrix}^{-1}$$

- **Theorem (Inverse of a Partitioned Matrix).** Given a matrix $A \in \mathbb{R}^{n \times n}$, two vectors $b, c \in \mathbb{R}^{n \times 1}$ and a number $d \in \mathbb{R}$ we have

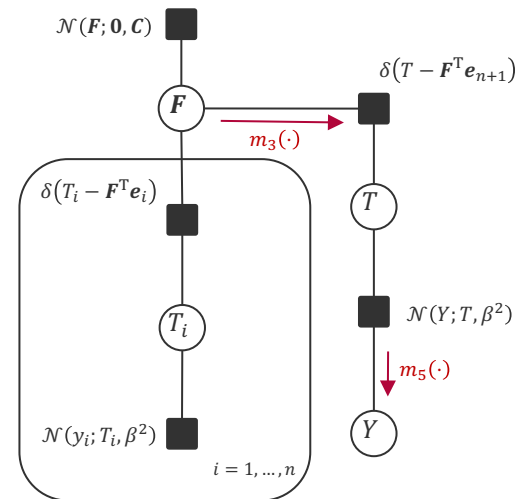
$$\begin{bmatrix} A & b \\ c^T & d \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} + e \cdot A^{-1} b c^T A^{-1} & -e \cdot A^{-1} b \\ -e \cdot c^T A^{-1} & e \end{bmatrix} \quad e = \frac{1}{d - c^T A^{-1} b}$$

■ Putting it all together

$$m_5(y) = \mathcal{N}(y; k^T (K + \beta^2 I)^{-1} y, \beta^2 + k - k^T (K + \beta^2 I)^{-1} k)$$

data uncertainty

model uncertainty



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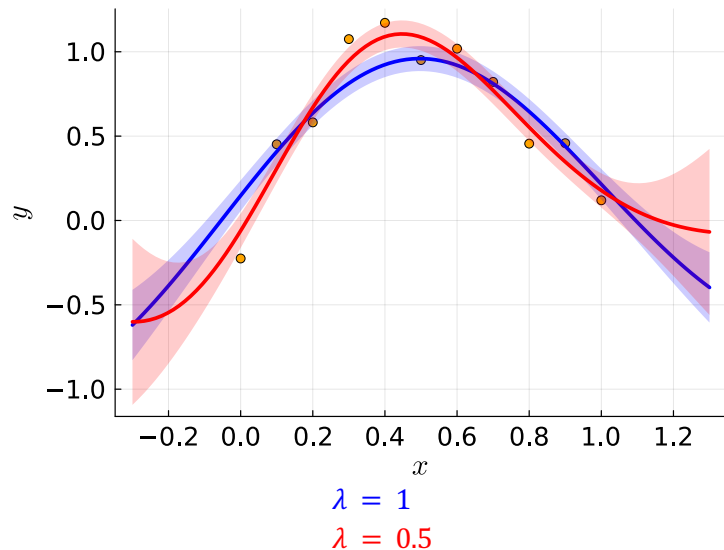
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Gaussian Process Regression: Example

$$m(x) = 0$$

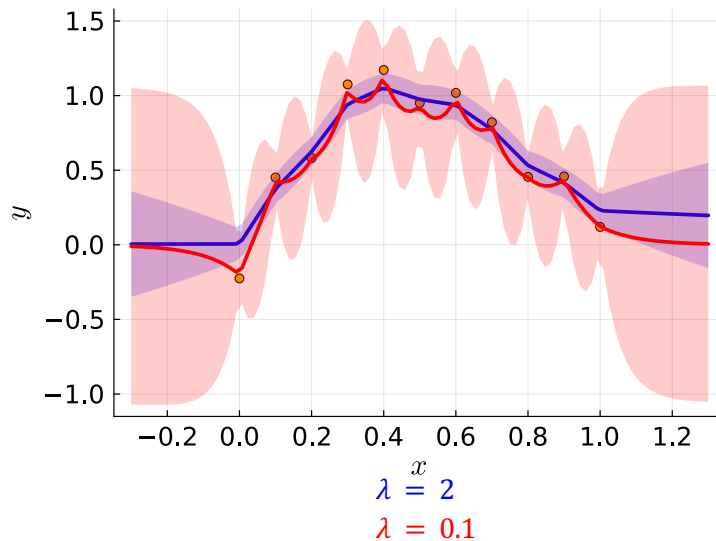
RBF Kernel

$$C(x, x') = \exp\left(-\frac{(x - x')^2}{\lambda^2}\right)$$



Ornstein-Uhlenbeck Kernel

$$C(x, x') = \exp\left(-\frac{|x - x'|}{\lambda}\right)$$



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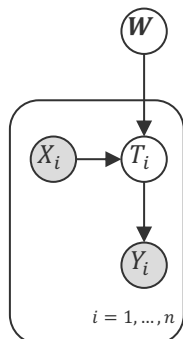
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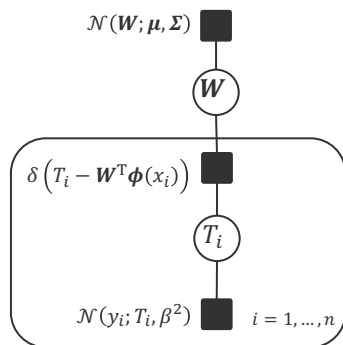
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Bayesian Linear Regression as Gaussian Process

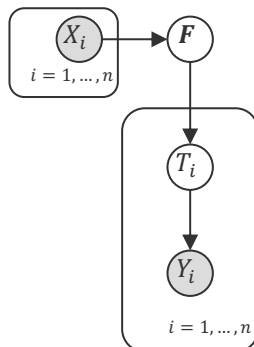
Bayesian Linear Regression



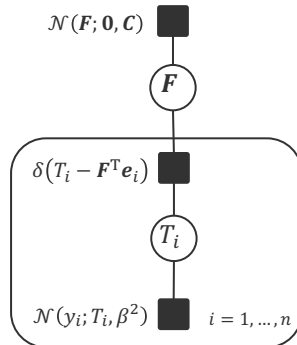
Training set



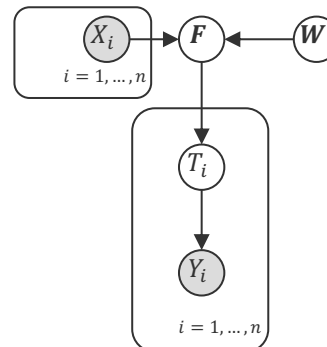
GP Regression



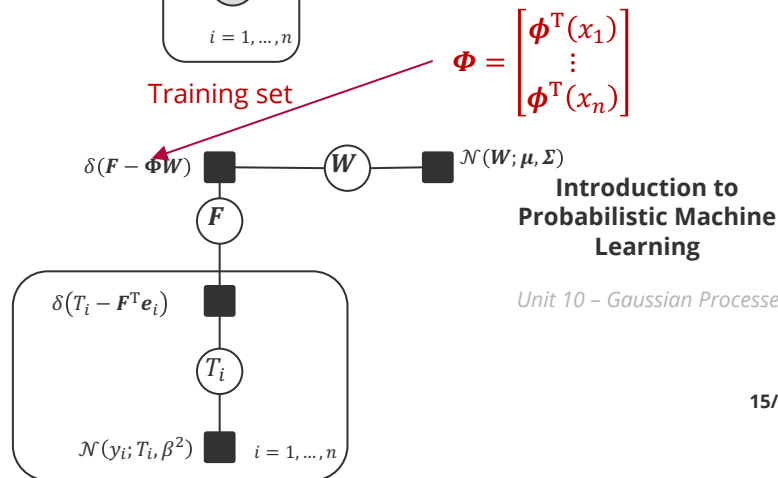
Training set



Bayesian Linear Regression (GP View)



Training set

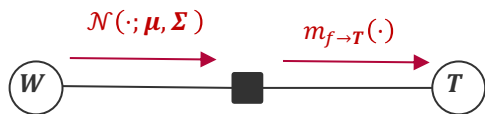


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Gaussian Processes: Weight Space View

Gaussian Projection Factor $\delta(F - \Phi W)$

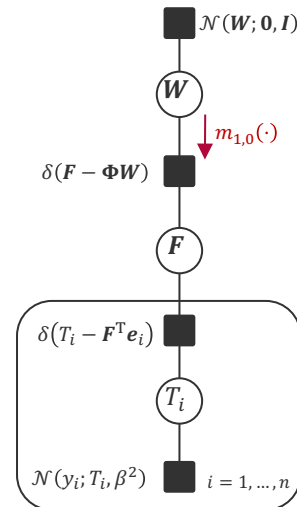


$$m_{f \rightarrow T}(t) = \int \delta(t - \Phi w) \cdot \mathcal{N}(w; \mu, \Sigma) dw = \mathcal{N}(t; \Phi \mu, \Phi \Sigma \Phi^T)$$

$$m_{1,0}(f) = \mathcal{N}(f; \mathbf{0}, \Phi \Phi^T) = \text{GP}(f; \mathbf{0}, \phi(x)^T \phi(x'))$$

$$\Phi = \begin{bmatrix} \phi^T(x_1) \\ \vdots \\ \phi^T(x_n) \end{bmatrix}$$

- **Equivalence:** A Gaussian Process is equivalent to the marginal likelihood of a linear basis function model with the prior $p(\mathbf{w}) = \mathcal{N}(\mathbf{w}; \mathbf{0}, \mathbf{I})$ and the covariance function $C(x, x') = \phi(x)^T \phi(x')$.



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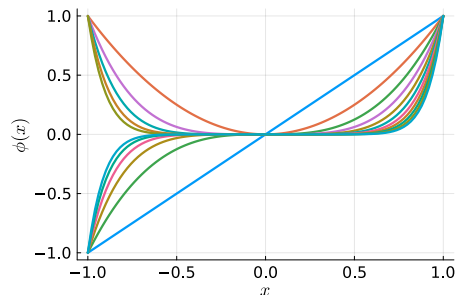
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	Number of Basis Functions	Computational Cost
Linear Basis Function Model	M	$O(M^3)$
Gaussian Process	∞	$O(n^3)$

Constructing Covariance Functions

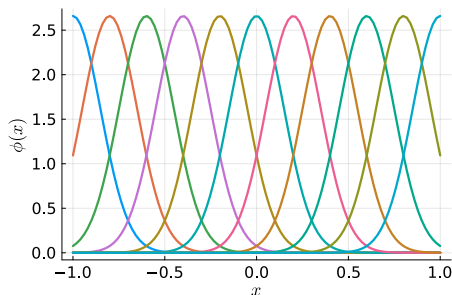
Polynomial Basis

$$\phi_j(x) = x^j$$



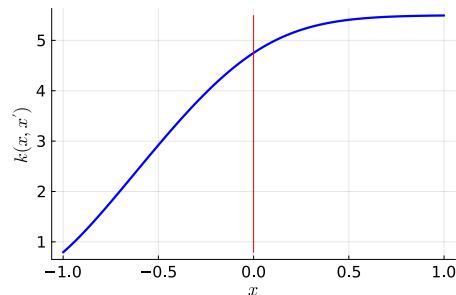
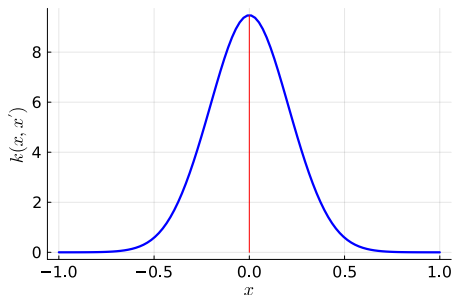
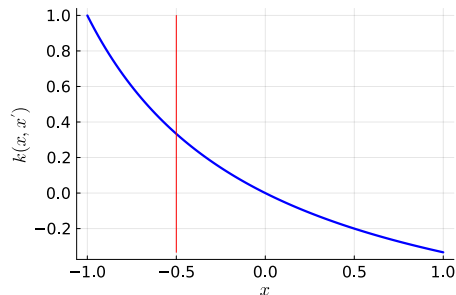
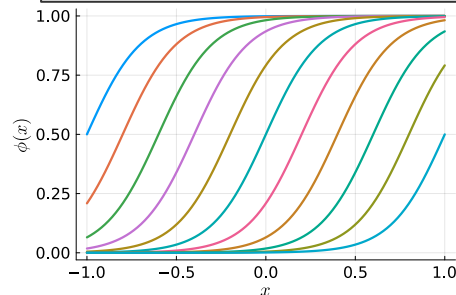
Gaussian Basis

$$\phi_j(x) = \mathcal{N}\left(x; \frac{j}{5} - 1, 0.15^2\right)$$



Sigmoid Basis

$$\phi_j(x) = \frac{\exp(x - 0.2j + 1)}{1 + \exp(x - 0.2j + 1)}$$



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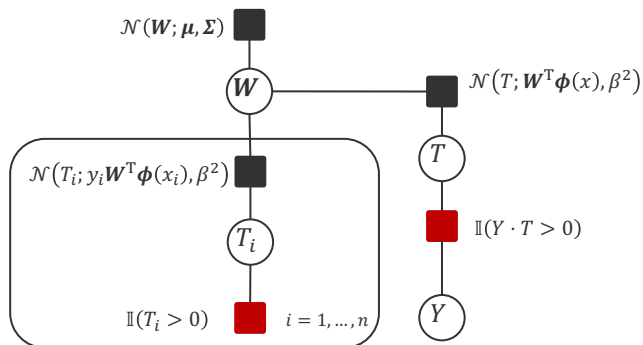
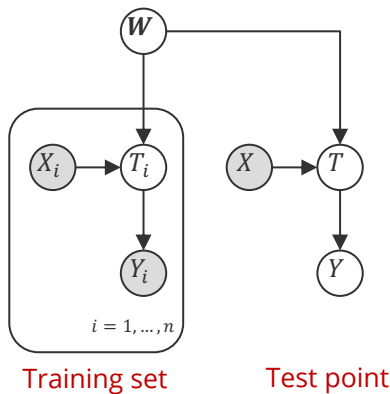
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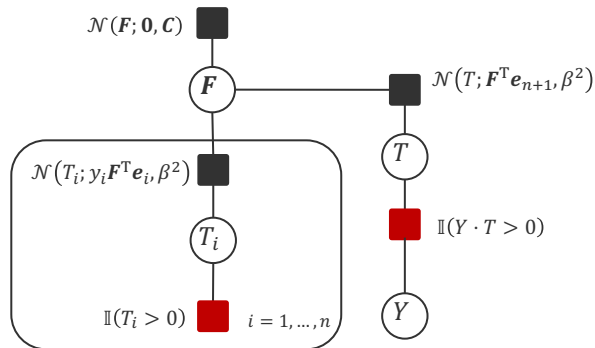
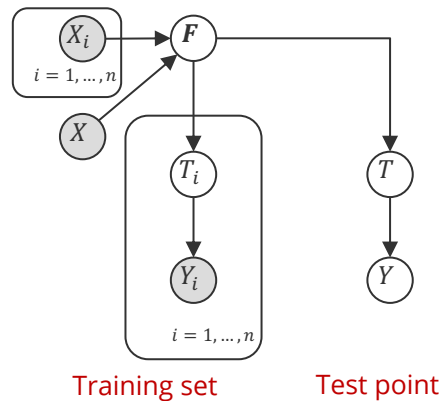
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Gaussian Process Classification as Factor Graph

Bayesian Linear Classification



Gaussian Process Classification



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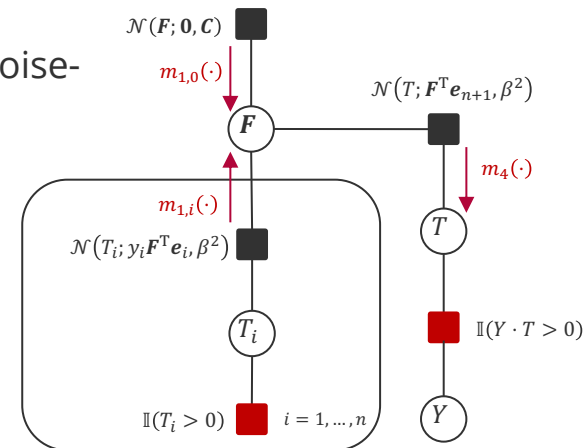
Gaussian Process Classification via Message Passing

- **Idea:** We approximate each incoming message from the Gaussian noise-projection factor $\mathcal{N}(T_i; y_i \mathbf{F}^T \mathbf{e}_i, \beta^2)$ with via $\mathcal{G}(\mathbf{f}; a_i y_i \mathbf{e}_i, b_i \mathbf{e}_i \mathbf{e}_i^T)$
 - GP prior message: $m_{1,0}(\mathbf{f}) = \mathcal{N}(\mathbf{f}; \mathbf{0}, \mathbf{C})$
 - Approximate data message: $m_{1,i}(\mathbf{f}) = \mathcal{G}(\mathbf{f}; a_i y_i \mathbf{e}_i, b_i \mathbf{e}_i \mathbf{e}_i^T)$
- Equivalent to Gaussian process regression with training points
 - At $(x_i, \frac{a_i}{b_i} y_i)$
 - And noise variance of $1/b_i$
- In this case, the message $m_4(\cdot)$ to T at a test point x is given by

$$m_4(t) = \mathcal{N}(t; \mathbf{k}^T \tilde{\mathbf{K}}^{-1} \tilde{\mathbf{y}}, \beta^2 + C(x, x) - \mathbf{k}^T \tilde{\mathbf{K}}^{-1} \mathbf{k})$$

$\mathbf{K} + \begin{bmatrix} 1 & \dots & 0 \\ \frac{1}{b_1} & \ddots & \vdots \\ 0 & \dots & \frac{1}{b_n} \end{bmatrix}$
 $\begin{bmatrix} a_1 y_1 \\ b_1 \\ \vdots \\ a_n y_n \\ b_n \end{bmatrix}$

model uncertainty
 data uncertainty



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Gaussian Process Classification via Message Passing

Given: n examples (x_i, y_i) where $y_i \in \{-1, +1\}$ and covariance function $C: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$

- 1. Initialize** $\mathbf{K}$ with $K_{i,j} = C(x_i, x_j)$, $\Sigma = \mathbf{K}$, $\mu = \mathbf{0}$ and $a_i = b_i = 0$
- 2. Repeat until** maximum change component-wise in $\mathbf{a}$ and $\mathbf{b}$ is small (e.g., $\leq 10^{-6}$)

For all $i = 1, \dots, n$

Step 1: Copy the i -th column of Σ into $\mathbf{z} = \Sigma_{:,i}$

Step 2: Compute $m = \frac{y_i \cdot \mu_i - a_i \cdot \Sigma_{i,i}}{1 - b_i \cdot \Sigma_{i,i}}$ and $s^2 = \frac{\Sigma_{i,i}}{1 - b_i \cdot \Sigma_{i,i}} + \beta^2$

Step 3: Compute $\tau = \frac{s \cdot V + m \cdot W}{s^2 \cdot (1 - W)}$ and $\rho = \frac{W}{s^2 \cdot (1 - W)}$ for $V = v\left(\frac{m}{s}\right)$ and $W = w\left(\frac{m}{s}\right)$
 Compute $a_i^{\text{new}} = \frac{\tau}{1 + \rho \cdot \beta^2}$ and $b_i^{\text{new}} = \frac{\rho}{1 + \rho \cdot \beta^2}$

Step 4: Compute $d = 1 + (b_i^{\text{new}} - b_i) \cdot \Sigma_{i,i}$

Update $\Sigma \leftarrow \Sigma - \frac{b_i^{\text{new}} - b_i}{d} \cdot \mathbf{z} \mathbf{z}^T$ and $\mu \leftarrow \mu + \frac{y_i \cdot (a_i^{\text{new}} - a_i) - \mu_i \cdot (b_i^{\text{new}} - b_i)}{d} \cdot \mathbf{z}$

Update $a_i \leftarrow a_i^{\text{new}}$ and $b_i \leftarrow b_i^{\text{new}}$

End For

- 3. Return** Cholesky decomposition $\mathbf{L}$ of $\mathbf{K} + \text{diag}\left(\begin{bmatrix} b_1^{-1} \\ \vdots \\ b_n^{-1} \end{bmatrix}\right)$ and $\alpha = \mathbf{L}^T \setminus \left(\mathbf{L} \setminus \begin{bmatrix} y_1 a_1 b_1^{-1} \\ \vdots \\ y_n a_n b_n^{-1} \end{bmatrix}\right)$

Majority of algorithm is additions, multiplication, and division of scalars!

Algorithm requires **no** matrix inverses – just matrix products!

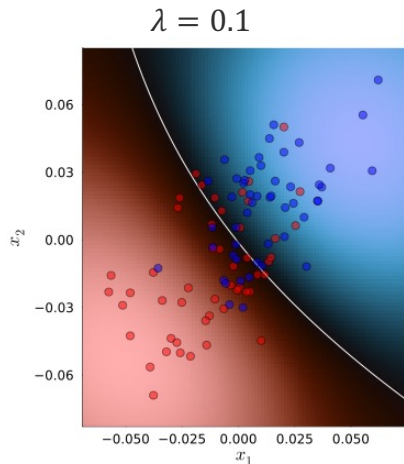
Predictions at x are done via a truncated Gaussian with

$$\mu = \mathbf{k}^T \alpha$$

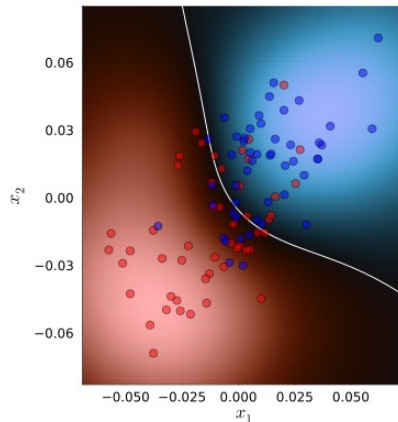
$$\sigma^2 = \beta^2 + C(x, x) - \mathbf{L}^T \setminus (\mathbf{L} \setminus \mathbf{k})$$

Gaussian Process Classification in Pictures

$n = 100$



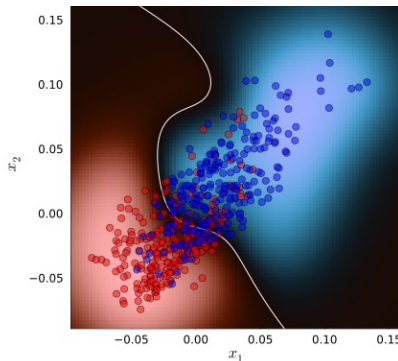
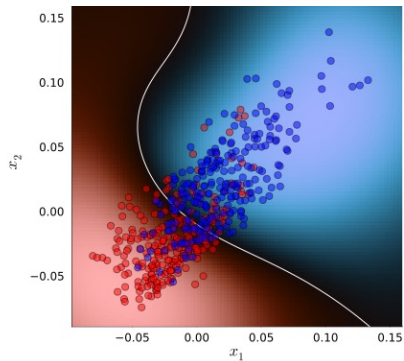
$\lambda = 0.03$



RBF Covariance Function

$$C(x, x') = \exp\left(-\frac{(x - x')^2}{\lambda^2}\right)$$

$n = 500$



**Introduction to
Probabilistic Machine
Learning**

Unit 10 – Gaussian Processes

1. Gaussian Processes

- Distribution over space of functions rather than function parameters
- Gaussian process can be modelled via a graphical model
- Gaussian process regression follows from standard message passing
- Gaussian Process models are equivalent to linear basis function models with a different parameterization (covariance functions instead of basis functions!)

2. Gaussian Process Classification

- Since the likelihood is no longer Gaussian, we approximate the posterior Gaussian process
- We can use the same approximate message passing (expectation propagation) as in Bayesian linear classification
- Many approximation schemes exist; for example the Laplace approximation (Kuss et al., 2005)

See you next week!